

Lab Exercise 3

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Section 1: Introduction to Plotly Express and Cufflinks

Importing Libraries

```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import cufflinks as cf
from plotly.offline import init_notebook_mode

init_notebook_mode.connected=True)
cf.go_offline()
```

```
gapminder = px.data.gapminder()
```

```
print(gapminder.head())
```

```
print(gapminder.shape)
```

```
print(gapminder.columns)
```

```
print(gapminder.info())
```

```
   country continent  year  lifeExp      pop  gdpPercap iso_alpha
0  Afghanistan    Asia  1952   28.801  8425333  779.445314      AFG
1  Afghanistan    Asia  1957   30.332  9240934  820.853030      AFG
2  Afghanistan    Asia  1962   31.997  10267083  853.100710      AFG
3  Afghanistan    Asia  1967   34.020  11537966  836.197138      AFG
4  Afghanistan    Asia  1972   36.088  13079460  739.981106      AFG

iso_num
0      4
1      4
2      4
3      4
4      4
(1704, 8)
Index(['country', 'continent', 'year', 'lifeExp', 'pop', 'gdpPercap',
      'iso_alpha', 'iso_num'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1704 entries, 0 to 1703
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   country    1704 non-null   object
1   continent  1704 non-null   object
2   year       1704 non-null   int64
3   lifeExp    1704 non-null   float64
4   pop        1704 non-null   int64
5   gdpPercap  1704 non-null   float64
6   iso_alpha  1704 non-null   object
7   iso_num    1704 non-null   int64
dtypes: float64(2), int64(3), object(3)
memory usage: 106.6+ KB
None
```

1. Load Gapminder Dataset

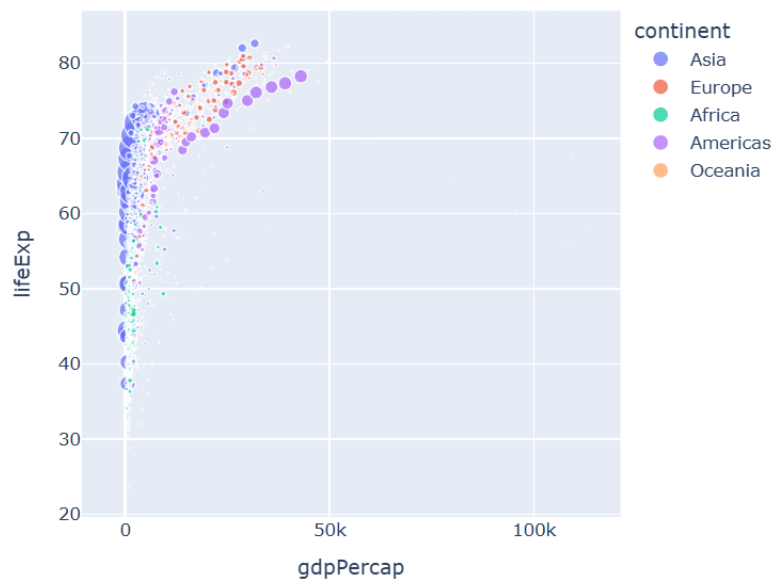
```
gapminder = px.data.gapminder()
print(gapminder.head())
print(gapminder.shape)
print(gapminder.columns)
print(gapminder.info())
```

2. Scatter Plot

```
fig = px.scatter(
    gapminder,
    x="gdpPerCap",
    y="lifeExp",
    color="continent",
    size="pop",
    hover_name="country",
    title="GDP Per Capita vs Life Expectancy"
)
```

```
fig.show()
```

GDP Per Capita vs Life Expectancy

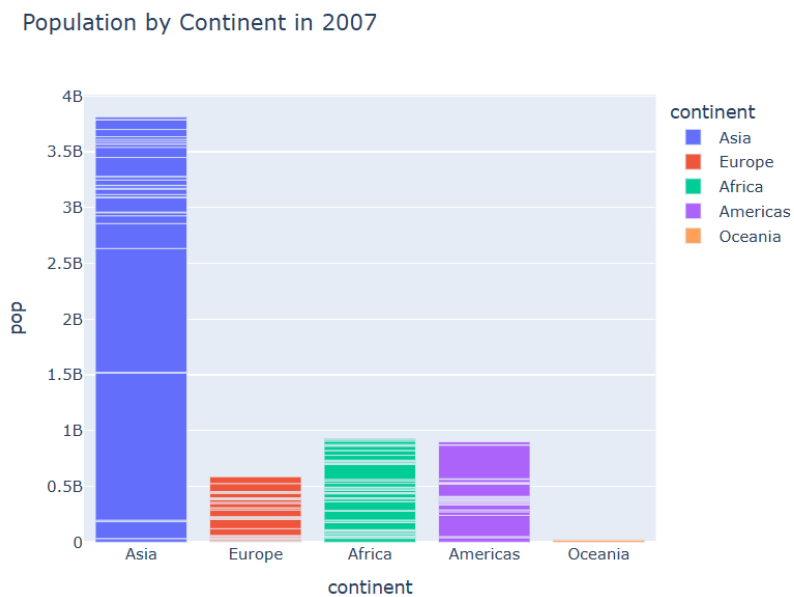


3. Bar Plot

```
df_2007 = gapminder[gapminder["year"] == 2007]
```

```
fig = px.bar(
    df_2007,
    x="continent",
    y="pop",
    color="continent",
    title="Population by Continent in 2007"
)
```

```
fig.show()
```



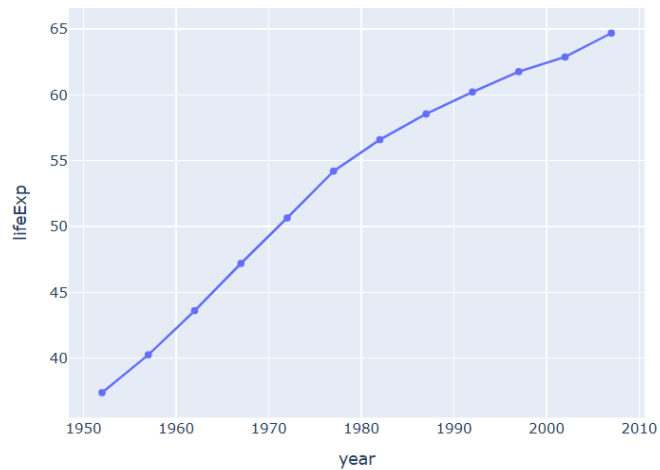
4. Line Plot

```
india = gapminder[gapminder["country"] == "India"]
```

```
fig = px.line(
    india,
    x="year",
    y="lifeExp",
    markers=True,
    title="Life Expectancy of India Over Time"
)
```

```
fig.show()
```

Life Expectancy of India Over Time

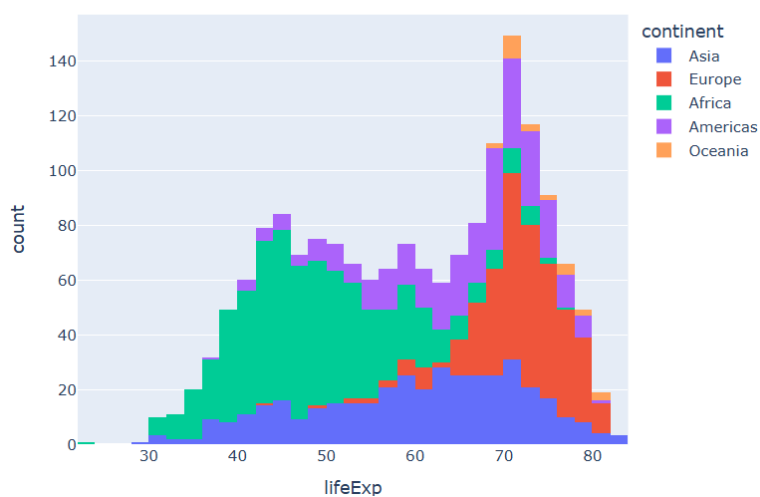


5. Histogram

```
fig = px.histogram(  
    gapminder,  
    x="lifeExp",  
    color="continent",  
    nbins=30,  
    title="Distribution of Life Expectancy"  
)
```

```
fig.show()
```

Distribution of Life Expectancy



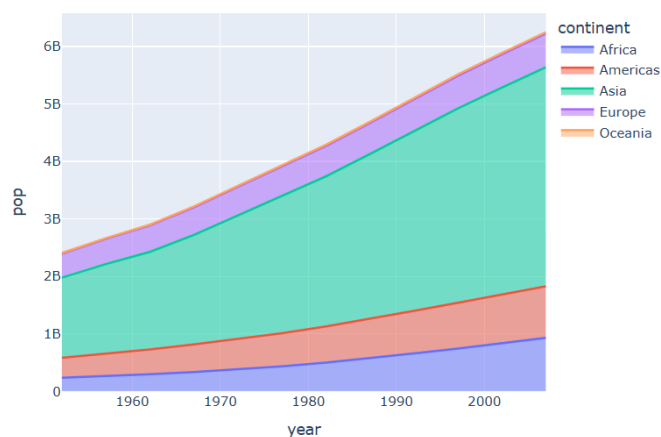
6. Area Chart

```
population_continent = gapminder.groupby(  
    ["year", "continent"],  
    as_index=False  
)["pop"].sum()
```

```
fig = px.area(  
    population_continent,  
    x="year",  
    y="pop",  
    color="continent",  
    title="Population Growth by Continent"  
)
```

```
fig.show()
```

Population Growth by Continent



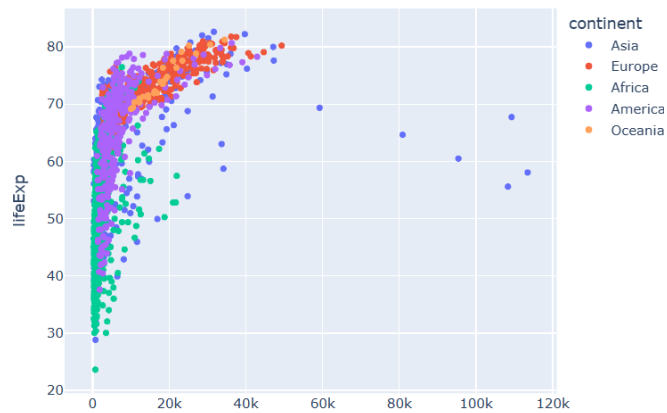
Section 2 : Interactive Visualizations

7. Hover Information

```
fig = px.scatter(  
    gapminder,  
    x="gdpPercap",  
    y="lifeExp",  
    color="continent",  
    hover_name="country",  
    hover_data=["year", "pop"],  
    title="Interactive GDP and Life Expectancy"  
)
```

```
fig.show()
```

Interactive GDP and Life Expectancy

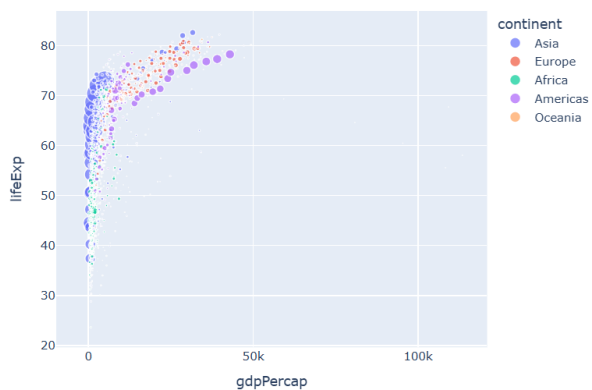


8. Color And Size

```
fig = px.scatter(  
    gapminder,  
    x="gdpPercap",  
    y="lifeExp",  
    color="continent",  
    size="pop",  
    hover_name="country",  
    title="GDP vs Life Expectancy"  
)
```

```
fig.show()
```

GDP vs Life Expectancy

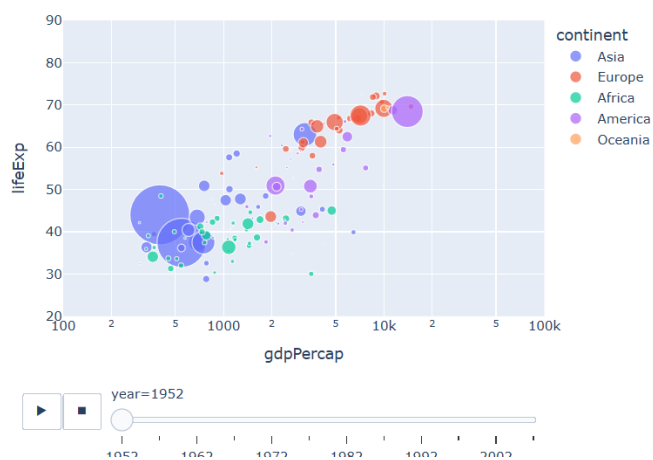


9. Animated Bubble Plot

```
fig = px.scatter(  
    gapminder,  
    x="gdpPercap",  
    y="lifeExp",  
    animation_frame="year",  
    animation_group="country",  
    size="pop",  
    color="continent",  
    hover_name="country",  
    log_x=True,  
    size_max=60,  
    range_x=[100, 100000],  
    range_y=[20, 90],  
    title="Gapminder Animated Bubble Plot"  
)
```

fig.show()

Gapminder Animated Bubble Plot



10. Interactive Dashboard — Statistical Plots

```
df_2007 = gapminder[gapminder["year"] == 2007]
```

```
fig = px.scatter(  
    df_2007,  
    x="gdpPercap",  
    y="lifeExp",  
    size="pop",  
    color="continent",  
    hover_name="country",
```

```

    title="Gapminder Dashboard"
)

```

```

fig.show()

```

Gapminder Dashboard



Section 3: Geographical Visualizations

11. World Choropleth using Plotly Express

```

df_2007 = gapminder[gapminder["year"] == 2007]

```

```

fig = px.choropleth(
    df_2007,
    locations="iso_alpha",
    color="lifeExp",
    hover_name="country",
    color_continuous_scale="Viridis",
    title="World Life Expectancy - 2007"
)

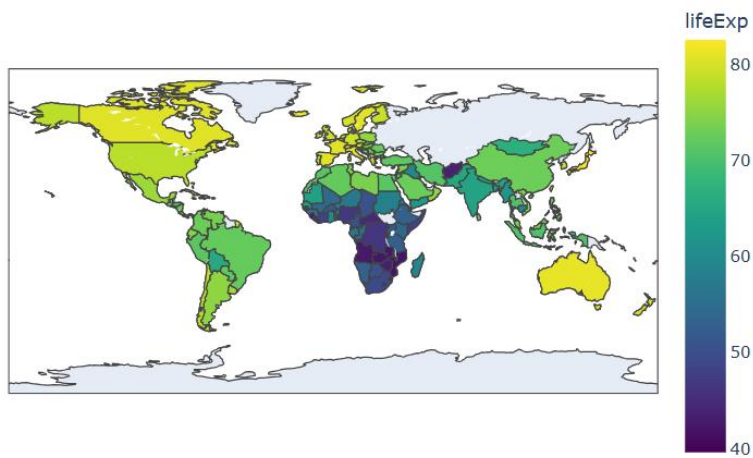
```

```

fig.show()

```

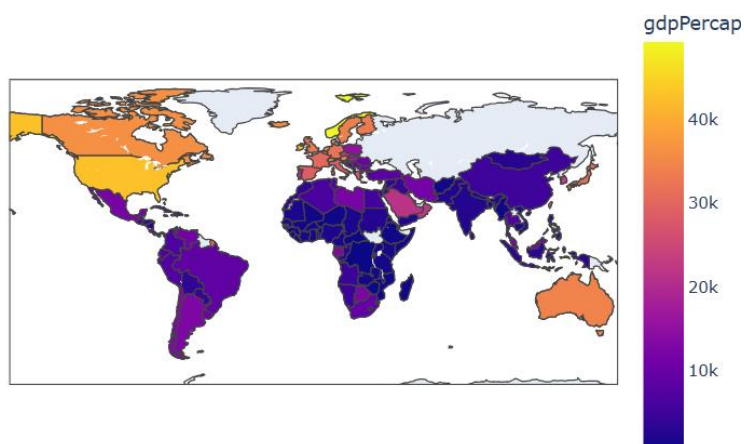
World Life Expectancy - 2007



12. World GDP Choropleth

```
fig = px.choropleth(  
    df_2007,  
    locations="iso_alpha",  
    color="gdpPercap",  
    hover_name="country",  
    color_continuous_scale="Plasma",  
    title="World GDP Per Capita - 2007"  
)
```

```
fig.show()  
World GDP Per Capita - 2007
```



13. Different Map Projection — Mercator

```
fig = px.choropleth(  
    df_2007,  
    locations="iso_alpha",  
    color="gdpPercap",  
    hover_name="country",  
    color_continuous_scale="Plasma",  
    title="World GDP Per Capita - 2007"  
)
```

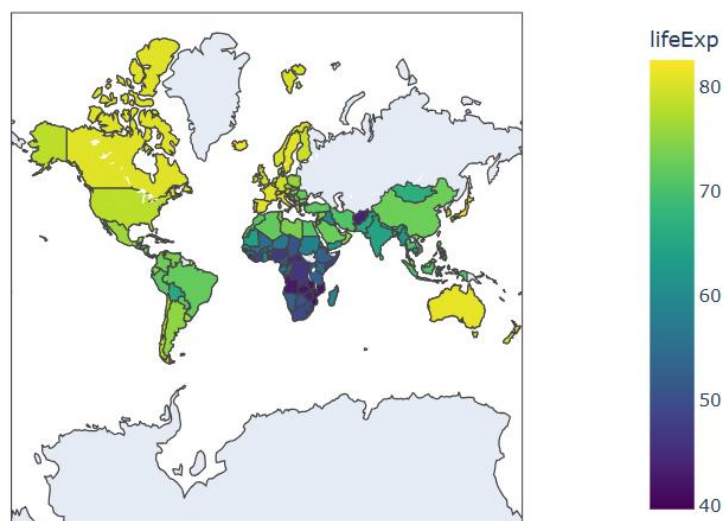
```

df_2007,
locations="iso_alpha",
color="lifeExp",
hover_name="country",
color_continuous_scale="Viridis",
projection="mercator",
title="Life Expectancy - Mercator Projection"
)

```

```
fig.show()
```

Life Expectancy - Mercator Projection



14. Orthographic Projection

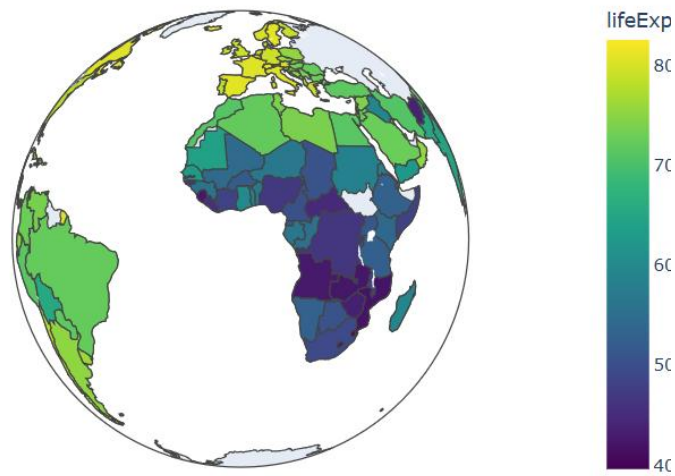
```

fig = px.choropleth(
    df_2007,
    locations="iso_alpha",
    color="lifeExp",
    hover_name="country",
    color_continuous_scale="Viridis",
    projection="orthographic",
    title="Life Expectancy - Orthographic Projection"
)

```

```
fig.show()
```

Life Expectancy - Orthographic Projection



15. Choropleth using Graph Objects

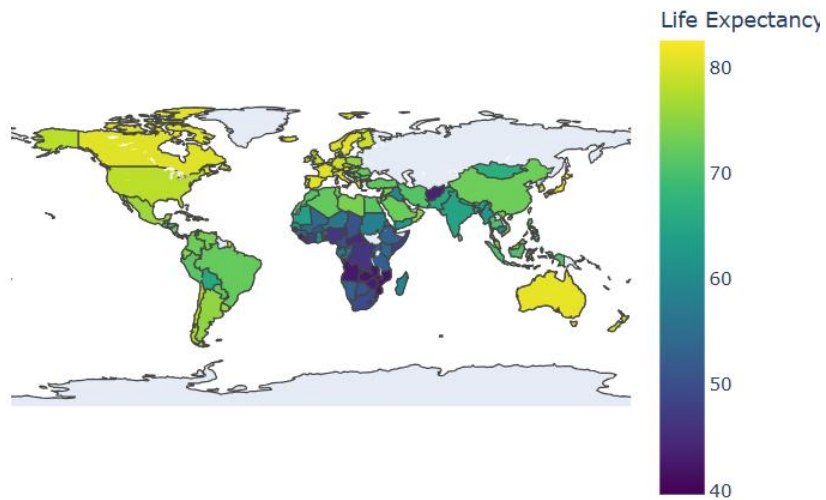
```
df_2007 = gapminder[gapminder["year"] == 2007]
```

```
fig = go.Figure(  
    data=go.Choropleth(  
        locations=df_2007["iso_alpha"],  
        z=df_2007["lifeExp"],  
        text=df_2007["country"],  
        colorscale="Viridis",  
        colorbar_title="Life Expectancy"  
    )  
)
```

```
fig.update_layout(  
    title_text="World Life Expectancy - 2007",  
    geo=dict(  
        showframe=False,  
        showcoastlines=True  
    )  
)
```

```
fig.show()
```

World Life Expectancy - 2007

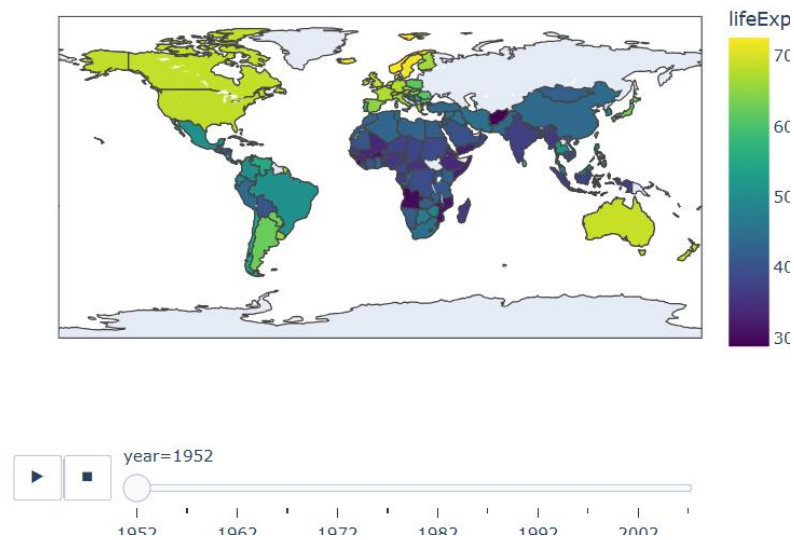


Section 4: Advanced Applications

16. Animated Geographical Visualization

```
fig = px.choropleth(  
    gapminder,  
    locations="iso_alpha",  
    color="lifeExp",  
    hover_name="country",  
    animation_frame="year",  
    color_continuous_scale="Viridis",  
    title="World Life Expectancy Over Time"  
)  
  
fig.show()
```

World Life Expectancy Over Time

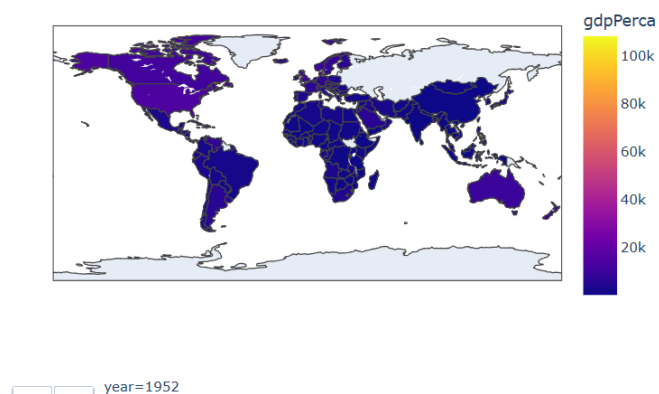


17. Animated GDP Map

```
fig = px.choropleth(
    gapminder,
    locations="iso_alpha",
    color="gdpPerCap",
    hover_name="country",
    animation_frame="year",
    color_continuous_scale="Plasma",
    title="World GDP Per Capita Over Time"
)
```

```
fig.show()
```

World GDP Per Capita Over Time



18. Bubble Plot with Hover Data

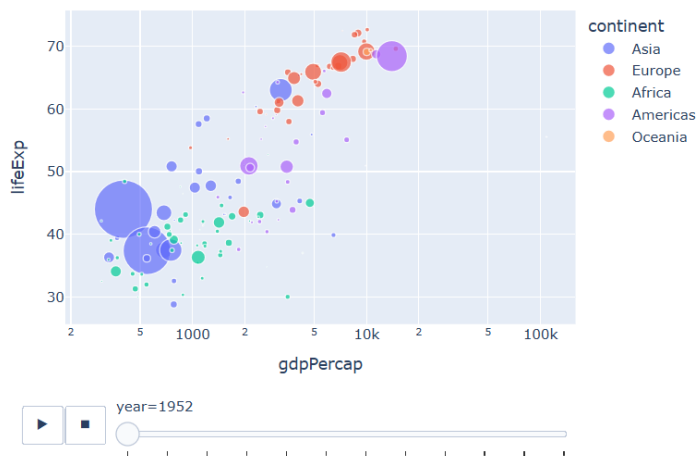
```

fig = px.scatter(
    gapminder,
    x="gdpPercap",
    y="lifeExp",
    color="continent",
    size="pop",
    animation_frame="year",
    hover_name="country",
    hover_data=["country", "continent", "year"],
    log_x=True,
    size_max=55,
    title="Population, GDP and Life Expectancy"
)

```

```
fig.show()
```

Population, GDP and Life Expectancy



19. Interactive Country Selection

```

selected = gapminder[
    gapminder["country"].isin(
        ["India", "China", "Japan", "United States", "Germany"]
    )
]

```

```

fig = px.line(
    selected,
    x="year",
    y="gdpPercap",
    color="country",
    markers=True,

```

```

    title="GDP Per Capita Comparison"
)

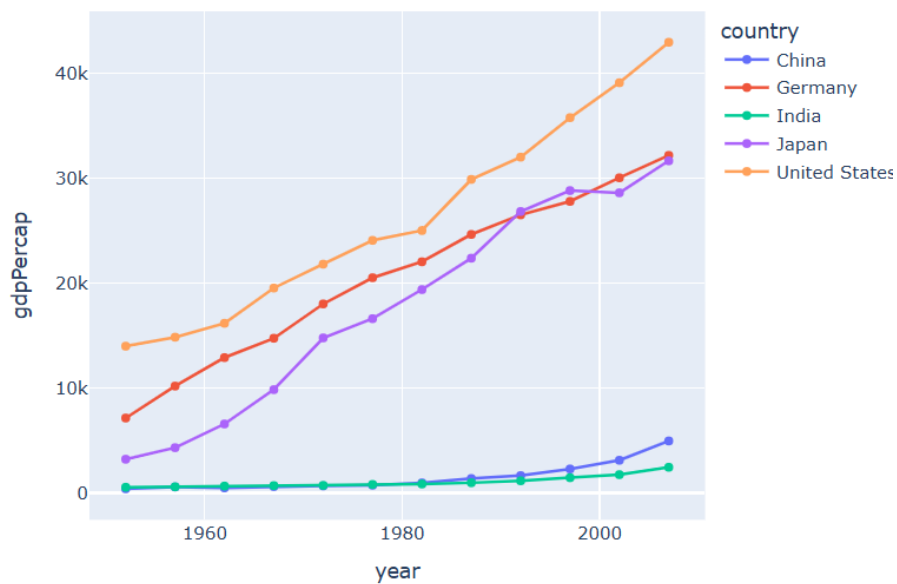
```

```

fig.show()

```

GDP Per Capita Comparison



20. Statistical + Geographical Visualization

Statistical

```

df_2007 = gapminder[gapminder["year"] == 2007]

```

```

fig1 = px.scatter(
    df_2007,
    x="gdpPerCap",
    y="lifeExp",
    size="pop",
    color="continent",
    hover_name="country",
    title="Statistical Visualization"
)

```

```
fig1.show()
```

Statistical Visualization



Geographical

```
fig2 = px.choropleth(  
    df_2007,  
    locations="iso_alpha",  
    color="lifeExp",  
    hover_name="country",  
    color_continuous_scale="Viridis",  
    title="Geographical Visualization"  
)  
fig2.show()
```

Geographical Visualization

